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Preference-Based Assessments

A Direct Comparison Between Discrete Choice With Duration and Composite Time Trade-Off Methods: Do They Produce Similar Results?

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ABSTRACT

Objectives: Discrete choice experiments including a duration attribute (DCEd) represent a promising candidate method for valuing health-related quality-of-life instruments. However, it has not been established that DCEd can produce similar results as composite time trade-off (cTTO) or EuroQol Valuation Technology (EQ-VT) valuations of the EQ-5D-5L instrument. This study provides a direct comparison between cTTO and EQ-VT, and DCEd valuation methods.

Methods: An EQ-VT study was conducted in Trinidad and Tobago to value the EQ-5D-5L. 1079 respondents each completed 10 cTTO tasks and 12 discrete choice experiments tasks without a duration attribute. A separate sample of 970 respondents each completed 18 split-triplet DCEd tasks. Several regression models were applied to the EQ-VT data, and the DCEd data were analyzed using mixed logit models with an exponential discount rate. The estimated values were compared using scatterplots and Bland-Altman plots.

Results: The ordering of dimensions was identical in level 5 for cTTO/EQ-VT and DCEd models, with pain/discomfort being the most important dimension and usual activities being least important. cTTO/EQ-VT models produced a value for state 55555 ranging between −0.52 and −0.69, whereas this was −0.543 for the nonlinear mixed logit model for the DCEd data. Scatterplots and Bland-Altman plots suggested excellent agreement between cTTO/EQ-VT and DCEd-based estimates.

Conclusions: CTTO/EQ-VT and DCEd valuations produce similar results when correcting DCEd for nonlinear time preferences. The ordering of importance of the dimensions and scale are identical, suggesting that the 2 methods measure the same construct and produce similar results.

Keywords: composite time trade-off, discrete choice experiment duration, discrete choice experiment, discrete choice with duration.

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Highlights

- Discrete choice experiments (DCEs) with duration is a promising method for health-state valuation. However, it is unknown whether it produces similar results to commonly used valuation methods.
- The current study provides a direct comparison of DCE with a duration attribute, and composite time trade-off and the EuroQol Valuation Technology valuation approach, showing excellent agreement between the methods, suggesting that all of these methods measure the same construct.
- Based on the results of this study, it may be possible to use DCE with a duration attribute interchangeably with composite time trade-off or EuroQol Valuation Technology. Furthermore, this study confirms the necessity of correcting for nonlinear time preferences in DCEs with a duration attribute.

Introduction

Health utilities, or health-state values are widely used in economic evaluations in healthcare, in which they are used as quality weights for the computation of quality-adjusted life-years (QALYs).[1] QALYs combine quality of life and duration of life into a single metric, measured on a scale anchored at 0 (dead) and 1 (full health). The value assigned to life-years is usually assumed to be linear. Information on health states is usually obtained by measuring the health-related quality of life (HRQoL) of a person, using instruments such as the EQ-5D-5L or SF-6D.[2,3] For example, the EQ-5D-5L instrument measures HRQoL on 5 dimensions: mobility, self-care, usual activities, pain/discomfort, and anxiety/depression. Respondents can indicate whether they have no problems, slight problems, moderate problems, severe problems, or extreme problems on each of these dimensions. The scores on each of the dimensions can then be summarized using a 5-digit score, representing the level of problems on each of the 5 dimensions, in order of presentation. For example, state 43125 would be a health state in which a person has severe problems (level 4) with walking about, moderate problems (level 3) with self-care, no problems (level 1) with usual activities, slight (level 2) pain or discomfort, and is extremely (level 5) anxious or depressed. These 5-digit health states can then be assigned a value using a value set or tariff, which serves as the HRQoL weight in the QALY model.

Value sets are often generated by collecting preferences for health states from members of the general population, using valuation methods such as the composite time trade-off (cTTO), Standard Gamble, and discrete choice experiments (DCEs).[4-6] For valuing EQ-5D-5L, a standardized protocol was developed: the EQ-VT protocol.[7,8] The EQ-VT protocol requires researchers to collect cTTO and DCE data via personal interviews using standardized methods. More than 35 value sets have been developed for the EQ-5D-5L using this protocol, as well as for other instruments.[9-14] Although the standardized EQ-VT protocol may bring several advantages, the EQ-VT protocol is incredibly costly, and it is time consuming to generate a value set because collecting cTTO data requires personal interviews, which may restrict countries in their capacity to develop a value set because of resource constraints. Although options for collecting cTTO with fewer resources have been explored, eg, by collecting cTTO data unsupervised via online surveys, this approach seems intractable and may lead to low-quality data with low face validity.[15] This questions the feasibility of conducting such studies without the presence of an interviewer. Unsupervised valuation tasks, such as DCEs, suffer less from such issues, although data quality problems can still be present, for which several solutions exist.[16,17]

The DCE as used in the EQ-VT protocol does not take into account duration of life, and values obtained via these DCE tasks are estimated on a latent scale, which is unsuitable for QALY computations. However, other studies have investigated the use of DCE with a duration attribute.[18] Although it was feasible to estimate values using these methods, the results were extreme, with the majority of health states receiving negative values in some studies.[19-22] Subsequent research showed that this was likely caused by the treating time preferences as linear, and taking into account nonlinearity in time preferences using discounting may solve this problem, leading to more reliable estimates.[23,24] The DCEs with duration methods using split-triplet designs seem to be promising candidates for future valuation studies for instruments such as the EQ-5D-5L.[23] However, these new DCEs with duration methods have never been directly compared with currently used valuation methods, such as the EQ-VT protocol.

The aim of the current study is to directly compare the results of an EQ-VT EQ-5D-5L valuation study, with a DCE study with a duration attribute (henceforth DCEd), both conducted in Trinidad and Tobago, with the aim of assessing whether DCEd produces similar results as cTTO. Here, the cTTO can optionally be supplemented with DCEs without duration data, as in the EQ-VT protocol. We formulate the following research question: how do values estimated via DCEd compare with values estimated using cTTO as used in the EQ-VT protocol? Sub aims of the study include a comparison of linear and nonlinear duration models for the DCEd, as well as a comparison with mean observed cTTO responses.

Methods

In this study, we conducted an EQ-VT study (reported fully in Bailey et al)[25] and a DCEd experiment for the EQ-5D-5L, allowing for a head-on comparison between 2 samples of the general public of Trinidad and Tobago. Ethical approval for this study was obtained from The University of The West Indies (Exemption letter CREC-SA.1468/03/2022 dated 7 March 2022).

Description of EQ-VT Data and Models

A target sample of 1000 respondents, representative of Trinidad and Tobago in terms of age, sex, and geography, was collected via face-to-face interviews, using the EQ-VT protocol. The EQ-VT survey was programmed in Limesurvey and each respondent completed a set of 10 cTTO tasks and 12 DCE tasks. In the cTTO task, respondents were asked to choose between a life of 10 years in some EQ-5D-5L health state or a shorter life in full health. Depending on the answer of the respondent, the number of life-years was subsequently varied, until the respondent indicated that he/she is indifferent, ending the task. Values were inferred from the number of life-years traded to avoid the health state in impaired health. For example, if a respondent was indifferent between 3 years in full health, or 10 years in state 12345, then the value assigned to this state is $\frac{3}{10} = 0.3$. More information on the cTTO method can be found in Janssen et al[5] (2013). After completing the cTTO tasks, each respondent was presented 12 DCE paired comparisons, in which the respondents were asked to choose between 2 EQ-5D-5L health states (with no duration specified).

For the cTTO tasks, each respondent was assigned a block of 10 health states, from a set of 10 blocks of a total of 86 unique health states, using the same design as used in most EQ-VT studies.[26] Each respondent was presented 1 mild health state (with level 2 problems on 1 dimension and no problems on all other 4 dimensions), state 55555 and 8 other states. For the DCE, an efficient design was used, comprising 20 blocks of 12 choice pairs. These choice pairs were designed to have level overlap on 2 dimensions; on 2 of the EQ-5D-5L dimensions, each choice pair would report the same level of problems, depicted in bold font, which simplified the task for the respondents and reduced attribute nonattendance.[27,28]

Interviewers were trained by 2 members of the study team (H.B. and B.R.) in a 4-day session, which included roleplay with the trainers and practice interviews with each other, friends, and family. Respondents were sampled via a panel company and interviewed in their own homes. The EQ-VT quality control protocol was applied to ensure the quality of the collected data.[29] Further details about the interview procedure and about the data collection for the EQ-VT can be found in Stolk et al[7] and Bailey et al.[25]

DCEd Data Collection

A target sample size of 1000 respondents was set, with each respondent completing a set of 18 split-triplet choice tasks. These split triplets were designed as in, eg, Jonker et al[21] and Jonker et al,[23] and separate choices between different health problems from choices between quantity and quality of life. Other operationalization of DCEd show large shares of respondents using additive decision rules, which is undesirable when assuming the multiplicative QALY model holds.[30] Respondents were first presented with a choice between 2 lives: life A and life B, both being EQ-5D-5L health states, with a duration specified that is equal between the 2 alternatives. Regardless of the preference for life A or life B, the respondent was subsequently presented a choice between life B and a different life, life C, which is presented as being in full health but for a smaller number of life-years than the number of life-years presented in life B. These tasks are simple and similar to TTO tasks and reduces the risk that respondents use additive decision rules. For 3 out of the 18 split triplets, life C was defined as immediate death rather than a number of years in full health. In the A versus B comparison, the 2 EQ-5D-5L health states had overlapping severity levels on 2 dimensions, to simplify the task for the respondents. Screenshots of the choice tasks are presented in the Appendix Figures. 1-7 in Supplemental Materials found at https://doi.org/10.1016/j.jval.2024.05.016.

The data were collected via a digital survey, programmed in Limesurvey. Respondents were first presented an information sheet and were asked to provide informed consent to participate. Subsequently, respondents completed a short demographic survey, including a self-complete of the EQ-5D-5L. Respondents were presented with a set of example tasks, with explanations of the choice tasks. After completing the warmup tasks, each respondent was presented with 3 sets of 6 split triplets, alternated by 2 short sets of demographic questions, followed by a short debriefing questionnaire. The layout of the DCE duration task was similar to the layout used in the EQ-VT data collection. Respondents of the DCEd survey were given a small compensation worth about $3 for their participation.

Respondents were sampled based on age and gender. Recruitment took place via a panel company, which used a mixed recruitment strategy. Part of the respondents completed the survey at their own convenience via a survey link, which was shared by email. However, there were concerns that online sampling alone would not be sufficient to achieve the target sample size. Therefore, a group of recruiters also recruited potential respondents in public places, such as libraries and public transport hubs, where respondents completed the survey on a laptop provided by the recruiter. Respondents were only allowed to complete the survey on devices such as personal computers, laptops, and tablets, which have a sufficiently large screen to show all the screen elements of the choice task in a large enough size.

Respondents that spent <12.5 seconds per split-triplet on average were excluded from the sample to ensure that the quality of the data was up to standards. Similar cutoffs have been used in other DCE studies to value EQ-5D instruments, such as, eg, 8.3 seconds per DCE pair.[31] We used a stricter cutoff because respondents are already familiar with health state B in the B/C choice. Removing respondents by response time is a method often used in DCE's to ensure sufficient data quality.[16] Respondents are excluded based on metadata rather than their responses, which ensures that respondents that do not make a serious effort to complete the DCE task are excluded from the analytical sample. Furthermore, the data from surveys completed via the internet panel were compared with the data collected via respondents recruited to complete the survey in a public place. If the mean completion time for a single recruiter of respondents, operationalized as using the same Internet Protocol (IP) address, was at least 20% lower than that of the respondents that completed the survey at their own convenience, all the data from that IP address were removed. Sensitivity analyses were performed at the end of the data collection using shorter and longer response times as exclusion criteria to ensure that a robust cutoff was chosen.

Health-State Designs and Design Updates

Because the efficiency of DCEd designs is sensitive to the prior specification of the model parameters, a near-orthogonal DCEd design was used to obtain an initial set of conditional logit estimates, which were subsequently used to create a more efficient (Bayesian) DCE design using the Time Preference Corrected QALY Designs (TPC-QD) software package.[32] To maximize statistical efficiency, the health-state design was updated 3 times, each based on batches of 200 respondents, until the priors that were used to generate the design were deemed sufficiently similar to the updated model estimates.

Each of the developed designs consisted of 10 sub designs with 18 split triplets, of which 15 had a C option consisting of a number of years in full health and 3 being comparisons with immediate death. The durations of the health states consisted of all strictly positive integer values of 15 years and smaller and 6 months. Alternatives A and B were forced to have the same value on the duration attribute, whereas alternative C always had a lower value on the duration attribute compared with alternatives A and B.

Modeling EQ-VT and DCEd Data and Comparing Between the Estimates

The EQ-VT data were modeled in Stata 18; using a heteroskedastic Tobit model for the cTTO data, a mixed logit model for the DCE data, and a hybrid model combining both datatypes, correcting for heteroskedasticity using a Tobit link for the cTTO data. In each of these models, values were estimated as disutility weights for the 20 level-dimension combinations of the EQ-5D-5L. A more detailed description of the study design and modeling of the EQ-VT data can be read in Bailey et al.[25]

The DCEd data were analyzed using mixed logit models of Jonker et al.[23] We considered both specifications with linear time preferences (ie, specification 1) and nonlinear time preferences (ie, specification 2 of Jonker et al[23]). These models estimate values as utility weights for the 20 level-dimension combinations of the EQ-5D-5L. In addition, a dummy variable representing the weight assigned to "immediate death" was estimated to test whether the scale derived from direct elicitation of preferences for "immediate death" aligned with the scale derived from the comparisons with full health. This dummy variable is estimated independently of the other parameters. The OpenBUGS code used is placed in Supplementary Materials found at https://doi.org/10.1016/j.jval.2024.05.016.

The modeled values for the EQ-VT (cTTO heteroskedastic Tobit, rescaled DCE mixed logit and hybrid heteroskedastic Tobit) were compared with the values estimated in the mixed logit models (with linear and nonlinear model specifications for the duration attribute) for the DCEd data. Values for all possible 3125 EQ-5D-5L health states were compared using scatterplots and Bland-Altman plots. Furthermore, for each of the 86 health states included in the cTTO health-state design, the mean observed cTTO response was calculated. These 86 means were subsequently compared with the predicted values of the models estimated on the DCEd and EQ-VT data, again using scatter plots and Bland-Altman plots.

Results

The EQ-VT data were collected between June 2022 and September 2022. A team of 11 interviewers interviewed a total of 1079 respondents, roughly representative of the population of Trinidad and Tobago. The DCEd data were collected between November 2022 and June 2023. A sample of 1581 respondents completed the 18 split-triplet tasks. 611 respondents were excluded because they failed the quality control criteria, leaving a final sample of 970 respondents.

206 respondents that completed the survey online were excluded because they spent on average <12.5 seconds and therefore were considered speeders. In addition, 405 respondents were recruited in public places and263 (62.46%) of these respondents spent on average <12.5 seconds per split-triplet and were also considered a speeders. Moreover, 302 (74.6%) of these respondents always chose B or always chose C in the B/C comparison (a phenomenon called flatlining). Because the data collected via this mode lacked validity because of speeding and flatlining, all data collected through this method (IP addresses that were used at least 3 times) were removed, and the data collection through these means was discontinued after the pilot data collection, 61.7% (n = 377) of the excluded respondents flatlined on B or C in the B/C comparison, whereas this was 22.4% (n = 217) in the sample included for the final analysis. The sample characteristics of the DCEd sample, as well as the EQ-VT, are presented in Table 1. Females were overrepresented in the EQ-VT sample(χ^2 test, P = .038), as well as older respondents(χ^2 test, P = .005). The sample characteristics of the excluded sample are reported in the Appendix Table 1 in Supplemental Materials found at https://doi.org/10.1016/j.jval.2024.05.016.

The results of the DCEd models are reported in Table 2. Both the linear and nonlinear model produced consistent, significant estimates. In the linear model, mobility level 5 receives the largest weight: −0.524, with pain/discomfort level 5 receiving the second-largest weight: −0.507. In the nonlinear model, pain-discomfort level 5 receives a slightly larger weight than mobility level 5: −0.365 versus −0.361. The remaining level 5 parameters are ordered as anxiety/depression, self-care, and usual activities for both the linear and nonlinear model. Health state 55555 is assigned a value of −1.2136 in the linear model, whereas this is −0.543 in the nonlinear model.

For the EQ-VT data, the modeled results are reported in Table 3. All models produced consistent and significant estimates. In the cTTO heteroskedastic Tobit model, the mixed logit model for the DCE data and the hybrid heteroskedastic Tobit model, level 5 pain/discomfort receives the largest weight, with estimates ranging between 0.464 and 0.541. Level 5 mobility receives the second-largest weight, followed by level 5 for anxiety/depression, self-care, and usual activities. Appendix Figure 8 in Supplemental Materials found at https://doi.org/10.1016/j.jval.2024.05.016 plots the coefficients for the 3 EQ-VT models against the coefficients of the nonlinear DCEd model.

Figure 1 shows the values for all possible 3125 health states for the nonlinear model estimated on the DCEd data (vertical axis), plotted against the values for the cTTO heteroskedastic Tobit, the rescaled EQ-VT mixed logit model, and the hybrid heteroskedastic Tobit model. The values predicted by the nonlinear DCEd model showed high correlations with the modeled cTTO values (0.954), the rescaled EQ-VT mixed logit (0.973), and 2 hybrid models (0.973).

Figure 2 shows Bland-Altman plots, comparing the values predicted by same 3 models based on the EQ-VT data with the predictions of the nonlinear DCEd model. On the vertical axes, the differences between the EQ-VT estimates and the DCEd estimates are shown, whereas on the horizontal axes, the averages of the 2 estimates are shown. Differences between estimates are generally smaller than 0.2 in absolute terms, with some exceeding 0.2 for the cTTO heteroskedastic Tobit model but not for the other 2 models. The DCEd model predicts slightly higher values compared with the 3 other models, especially in the middle of the scale, as can be seen from the Bland-Altman plot. The mean differences between the DCEd estimates and the cTTO heteroskedastic Tobit, rescaled DCE and hybrid heteroskedastic Tobit were 0.041, 0.011, and 0.036, respectively. The mean absolute difference (MAD) between the DCEd estimates and estimates for the 3 EQ-VT models for the full set of 3125 states were 0.083, 0.051, and 0.060, respectively. In contrast, the MAD for the cTTO heteroskedastic Tobit and the rescaled DCE from the EQ-VT sample was 0.045.

When comparing the mean observed cTTO values with the modeled values for the DCEd, cTTO Tobit heteroskedastic, rescaled mixed logit, and the hybrid model, similar performance is observed between the methods, with the mean absolute error for the DCE duration being slightly larger at 0.081, compared with the cTTO (0.042), rescaled DCE (0.053), and hybrid (0.044) models (see Appendix Figs. 9 and 10 in Supplemental Materials found at https://doi.org/10.1016/j.jval.2024.05.016 for scatter and Bland-Altman plots of the mean observed cTTO values)

Sensitivity analyses were performed to test the robustness of the results. A Bland-Altman plot comparing the DCEd and EQ-VT using the linear model instead of the nonlinear model is reported in Figure 3. Scatterplots comparing these methods are reported in online supplemental Appendix Figure 11 in Supplemental Materials found at https://doi.org/10.1016/j.jval.2024.05.016. Furthermore, because we excluded respondents who spent <12.5 seconds from the analysis, we have also conducted sensitivity analyses using thresholds of 10 and 15 seconds to confirm the robustness of the results. These results are reported in Appendix Table 2 in Supplemental Materials found at https://doi.org/10.1016/j.jval.2024.05.016.

Discussion

Main Findings

This study shows that health-state valuations based on cTTO as used in the EQ-VT protocol and on DCEd can produce similar

Table 1. Sample characteristics.

Variables	DCEd, n (%)	EQ-VT, n (%)	Population (2011 census), %
Age (in years)			
18-24	156 (16.1)	163 (15.1)	15.40
25-34	230 (23.7)	234 (21.7)	23.00
35-44	181 (18.6)	220 (20.4)	18
45-54	187 (19.3)	156 (14.5)	18.40
55-64	126 (13)	175 (16.2)	13.30
>65	90 (9.3)	131 (12.1)	11.80
Gender			
Male	484 (49.9)	489 (45.3)	49.80
Female	486 (50.1)	590 (54.7)	50.20
Education			
Incomplete secondary	78 (8)	204 (18.9)	28.8
Complete secondary	358 (36.9)	469 (43.5)	51.5
Complete technical/vocational	132 (13.6)	139 (12.9)	8.2
Complete tertiary/university	402 (41.5)	267 (24.8)	11.5
Ethnicity			
Afro	457 (48.1)	369 (34.2)	37.1
Indo	162 (16.7)	438 (40.6)	40.0
Mixed/other	351 (36.2)	272 (25.2)	22.9

DCEd indicates discrete choice experiment including a duration attribute; EQ-VT, EuroQol Valuation Technology.

Table 2. Linear and nonlinear mixed logit estimates for the DCEd data, presented as utility weights for the level-dimension combinations of the EQ-5D-5L.

	Linear mixed logit			Nonlinear mixed logit		
coefficient	Beta	SD	*P* value	Beta	SD	*P* value
mo2	−0.073	0.0106	.000	−0.033	0.007	.000
mo3	−0.159	0.0149	.000	−0.095	0.009	.000
mo4	−0.323	0.0251	.000	−0.214	0.012	.000
mo5	−0.524	0.0391	.000	−0.361	0.017	.000
sc2	−0.075	0.0107	.000	−0.039	0.007	.000
sc3	−0.118	0.0129	.000	−0.072	0.008	.000
sc4	−0.272	0.0213	.000	−0.181	0.011	.000
sc5	−0.409	0.0307	.000	−0.283	0.014	.000
ua2	−0.051	0.0103	.000	−0.022	0.007	.007
ua3	−0.099	0.0124	.000	−0.058	0.009	.000
ua4	−0.206	0.0174	.000	−0.135	0.010	.000
ua5	−0.307	0.0232	.000	−0.210	0.012	.000
pd2	−0.091	0.0110	.000	−0.059	0.007	.000
pd3	−0.146	0.0135	.000	−0.098	0.008	.000
pd4	−0.323	0.0255	.000	−0.230	0.014	.000
pd5	−0.507	0.0386	.000	−0.365	0.019	.000
ad2	−0.090	0.0106	.000	−0.052	0.007	.000
ad3	−0.200	0.0167	.000	−0.126	0.009	.000
ad4	−0.373	0.0285	.000	−0.260	0.015	.000
ad5	−0.467	0.0360	.000	−0.324	0.017	.000
Dead	−3.235	0.4162	.000	−0.271	0.093	.005
Rate	-			0.235	0.009	.000
LL	−13 950			−12 650		
v(55555)	−1.214			−0.543		

Note. A separate parameter was estimated to estimate the location of dead in the dead triplets.
DCEd indicates discrete choice experiment including a duration attribute; LL, log-likelihood.

results when using nonlinear models for the DCEd data. The ordering of dimensions is comparable between the nonlinear DCEd model and the models estimated on the EQ-VT data. Furthermore, the scale produced by the nonlinear mixed logit model (ranging from 1 to −0.556) is similar to those of the models estimated on the EQ-VT data (all ranging from 1, to values for the pits state of −0.611, −0.686, and −0.563). All 3 models estimated on the EQ-VT data show high agreement with the DCEd estimates, as can be seen from the scatterplots, Bland-Altman plots and MAD between the methods. Furthermore, the estimated values for the DCEd model differ little from observed cTTO means.

When using a linear DCEd model, the ordering of the dimensions is the same as for the EQ-VT models, but the scale is very different.

Interpretation

The agreement between the cTTO/EQ-VT data and DCEd is high, as shown by the scatterplots, Bland-Altman plots, and MADs. Because the MAD for cTTO and latent-scale DCE was 0.045, the MADs for DCEd (cTTO 0.083, latent-scale DCE 0.051, and hybrid 0.060) can be considered small, especially given that the cTTO and latent-scale DCE were collected in the same sample, whereas the DCEd and EQ-VT data were not. Furthermore, Appendix Figure 8 in Supplemental Materials found at https://doi.org/10.1016/j.jval.2024.05.016 shows that the ordering of the dimensions, particularly for level 5, was similar between the DCEd and the other estimates. The size of the coefficient for pain/discomfort level 5 was smaller in the DCEd model compared with the other 3 models.

The results of the current study suggest that cTTO/EQ-VT and DCEd valuations lead to near-identical values, making DCEd a potentially cost-effective substitute for the current golden standard for valuing EQ-5D instruments, the EQ-VT protocol, and possibly for valuing other instruments as well. The ability to field DCEd online and unattended eliminates the need for expensive personal interviews, training, management, and extensive quality control procedures in comparison with EQ-VT studies.[29] However, because this study offers a single comparison within 1 country, further research is required to assess agreement in different language and cultural settings.

A relatively high annual discount rate of 23.5% (95% CI 21.7%-25.3%) was found in the current study, which is the population mean discount rate that best describes the data of the full sample.

Table 3. results analyses EQ-VT data; disutility weights for the 20 dimension-level combinations of the EQ-5D-5L, for selected models estimated on the EQ-VT data.

	TTO Heteroskedastic Tobit			Mixed logit latent scale DCE				Hybrid Heteroskedastic Tobit		
coefficients	Beta	SE	*P* value	Beta	SE	*P* value	Rescaled Beta	Beta	SE	*P* value
mo2	0.014	0.007	.048	0.7456	0.098	.000	0.0478	0.027	0.005	.000
mo3	0.058	0.013	.000	1.5087	0.123	.000	0.0967	0.085	0.006	.000
mo4	0.162	0.014	.000	3.1647	0.179	.000	0.2029	0.187	0.006	.000
mo5	0.357	0.014	.000	5.7013	0.298	.000	0.3655	0.368	0.007	.000
sc2	0.029	0.007	.000	0.4729	0.110	.000	0.0303	0.024	0.004	.000
sc3	0.074	0.011	.000	1.2360	0.111	.000	0.0792	0.072	0.006	.000
sc4	0.159	0.013	.000	2.4271	0.150	.000	0.1556	0.150	0.006	.000
sc5	0.221	0.012	.000	3.7728	0.197	.000	0.2418	0.232	0.006	.000
ua2	0.016	0.007	.021	0.1729	0.108	.109	0.0111	0.011	0.004	.007
ua3	0.087	0.011	.000	0.9613	0.117	.000	0.0616	0.065	0.006	.000
ua4	0.145	0.011	.000	2.1515	0.140	.000	0.1379	0.146	0.006	.000
ua5	0.216	0.013	.000	3.1860	0.176	.000	0.2042	0.219	0.006	.000
pd2	0.026	0.006	.000	1.2593	0.111	.000	0.0807	0.044	0.004	.000
pd3	0.102	0.013	.000	2.3727	0.139	.000	0.1521	0.128	0.006	.000
pd4	0.316	0.013	.000	5.0310	0.238	.000	0.3225	0.311	0.007	.000
pd5	0.541	0.016	.000	7.7300	0.340	.000	0.4955	0.480	0.008	.000
ad2	0.024	0.006	.000	0.2633	0.119	.027	0.0169	0.020	0.004	.000
ad3	0.057	0.012	.000	1.2371	0.137	.000	0.0793	0.074	0.006	.000
ad4	0.168	0.012	.000	2.4356	0.192	.000	0.1561	0.161	0.006	.000
ad5	0.272	0.011	.000	3.9882	0.246	.000	0.2556	0.264	0.006	.000
LL	−4590.88			7775.5				−12 835.86		
v(55555)	−0.611			−0.686				−0.563		

DCE indicates discrete choice experiment; EQ-VT, EuroQol Valuation Technology; LL, log-likelihood; TTO, time trade-off.

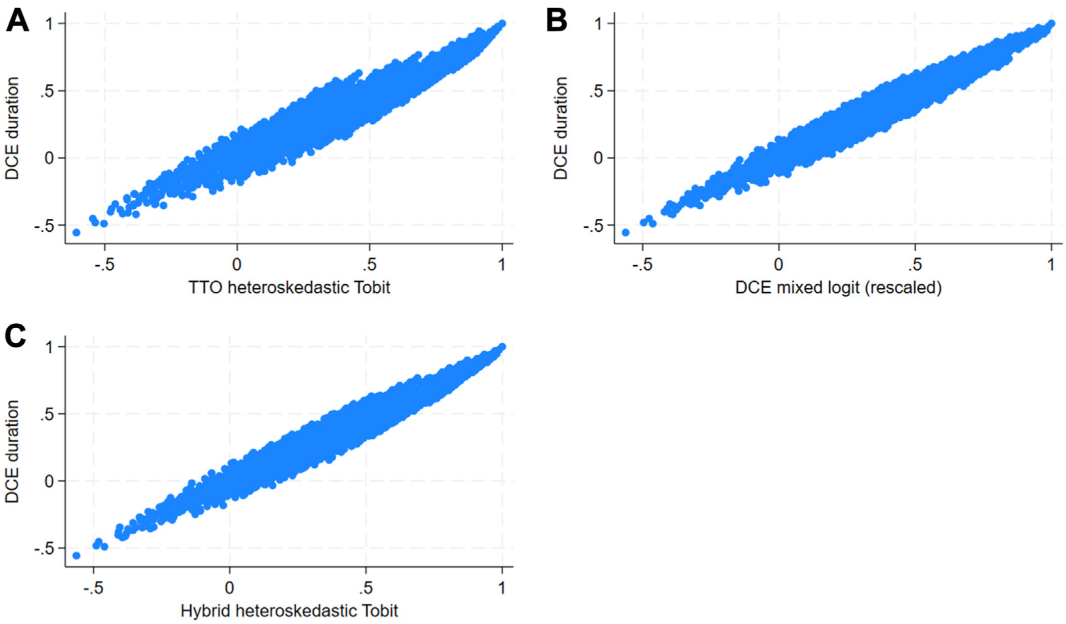
Previous studies produced lower discount rates. For example, Himmler et al[33] found a discount rate of 17.3% while valuing the Well-being Of Older People instrument, using the same methodology as in this study.[23,33] In the EQ-VT protocol, it is not possible to measure time preferences. Corrections for time preferences have been proposed.[34,35] Although it would be preferable to compare discounting-corrected cTTO values with discounting-corrected DCEd estimates, correcting for time preferences at the aggregate level in cTTO valuations has a smaller impact on values than in DCEd valuations.[36] In the DCEd method, discounting is used in a model-based extrapolation to anchor a latent scale onto the 0 (dead) 1 (full health) QALY scale. In the cTTO, this extrapolation is not needed because the responses are already anchored on the QALY scale, and corrections of TTO values are shown to have only a small impact.[34] The fact that the DCEd estimates were corrected for time preferences, and the EQ-VT data were not, is therefore unlikely to have a large impact on the comparison in the current article.

Discount rates in DCEd studies should be treated merely as a correction factor that disentangles time preferences from health-state preferences. In health technology assessment, only health-state preferences are taken into account, and discount rates vary between jurisdictions and are rarely based on empirical discount rates obtained from survey respondents. A review of discount practices by decision bodies shows that only in a few cases, empirical discount rates are used, and these rates are always financial discount rates, not discount rates for health-state preferences.[37] The discount rate therefore is merely a tool to correct health-state preferences and not important by itself. This study showed that using linear models in DCEd produces such extreme scales, which are unlikely to compare with EQ-VT data in any situation and should be disqualified as a feasible model. Although this study showed that within this study, EQ-VT and DCEd show very high agreement, more research is needed to ensure the generalizability of these findings. Different study settings (eg, differing quality of sampling via online panels between countries and self-selection of respondents for self-completion of a DCEd survey vs participation in EQ-VT interviews) may affect the results of these studies, which requires further investigation before DCEd can be used in parallel to EQ-VT. If DCEd performs well in such investigations, DCEd may be a cost-effective alternative to the EQ-VT and may help expand the number of available value sets for instruments such as the EQ-5D.

Limitations and Strengths

One of the limitations of the current study is that the methods used are not robust to data quality issues. In regular DCEs, statistical methods, such as the garbage class mixed logit model, can detect random response behavior, which is impossible in the current method.[17] Respondents consistently choosing a specific

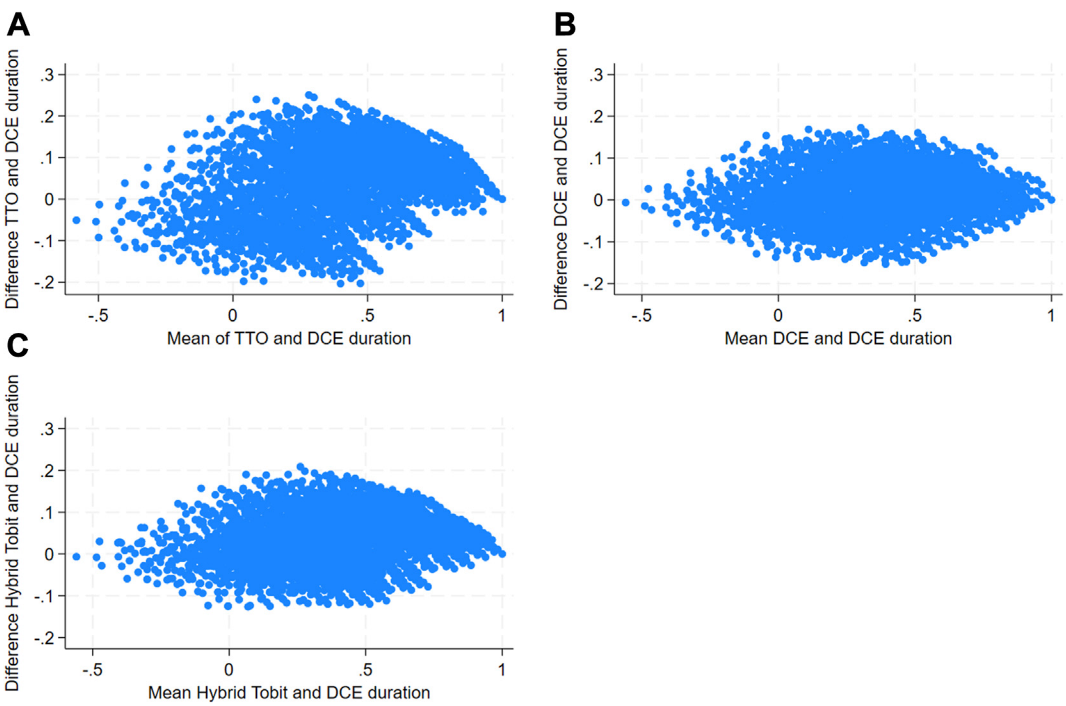
Figure 1. Scatterplot comparing the DCEd with EQ-VT using (A) A TTO tariff based on a heteroskedastic Tobit model. (B) Latent scale mixed logit model (rescaled to the value of 55555 in the hybrid Tobit model). (C) Hybrid heteroskedastic Tobit model.



DCEd indicates discrete choice experiments including a duration attribute; TTO, time trade-off.

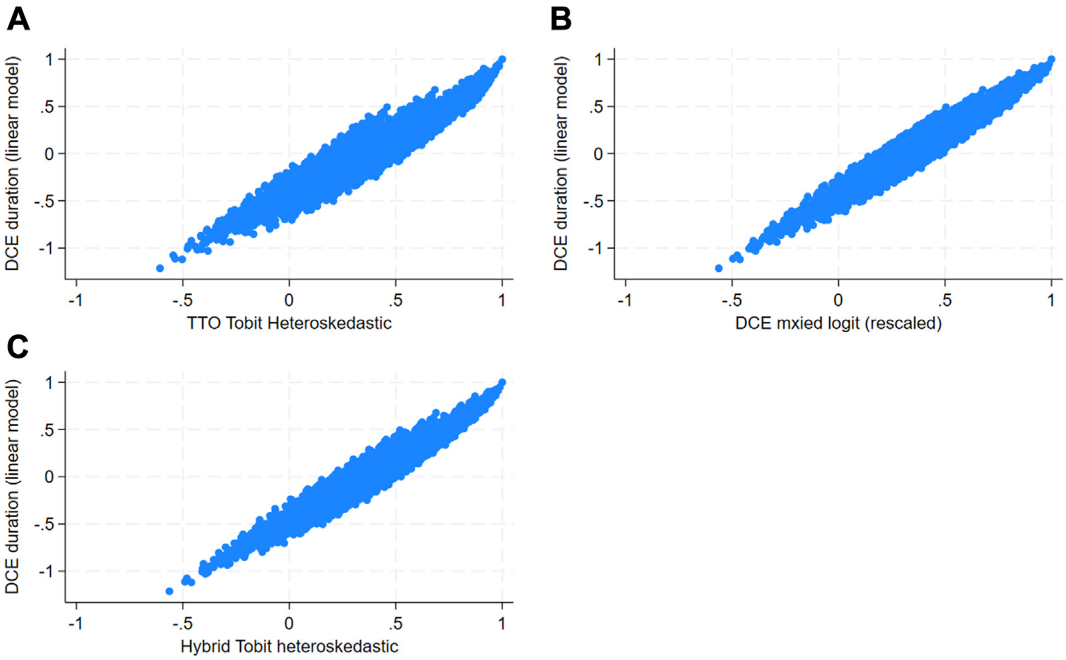
option in the “B”/”C” comparison may indicate either a strong preference for longevity or quality of life, or a lack of attention to the task, which poses a challenge because the model cannot distinguish between these possibilities. Moreover, the impact of these respondents is more significant in the current method, biasing scale and rate parameters, potentially invalidating the

Figure 2. Bland-Altman plots comparing the DCEd estimates with the EQ-VT data using (A) A TTO tariff based on a heteroskedastic Tobit model. (B) Latent scale mixed logit model (rescaled to the value of 55555 in the hybrid Tobit model). (C) Hybrid heteroskedastic model (D) Hybrid heteroskedastic Tobit model.



DCEd indicates discrete choice experiments including a duration attribute; TTO, time trade-off.

Figure 3. Scatter plots comparing the linear DCEd estimates with the EQ-VT data using: (A) a TTO tariff based on a heteroskedastic Tobit model. (B) latent scale mixed logit model (rescaled to the value of 55555 in the hybrid Tobit model). (C) hybrid heteroskedastic Tobit model.



DCEd indicates discrete choice experiments including a duration attribute; TTO, time trade-off.

models. The study observed this issue, especially with respondents rushing through the task, leading to poor modeling results. Removing speeders and discontinuing data collection in public places resolved this problem. A potential solution that is suggested is randomizing the order of presentation of choice alternatives to eliminate correlation with flatlining behavior, which requires further testing.

Another limitation is the disparity in demographic properties between the EQ-VT and DCE duration data samples, including the oversampling of females in the EQ-VT study and differences in education and ethnicity. However, these differences are considered unlikely to significantly affect estimated values. Additionally, the DCEd and EQ-VT data were collected in different samples, and there was a time lag between the collection of EQ-VT and DCE data, around 6 months, with expectations that any differences caused by this lag would be minor. Another limitation is that we had to cancel and discard in-person data collection entirely. This resulted in a substantial loss of sample, including respondents who were not identified as speeders. However, because over 60% of the sample failed this criterion, we could not confidently include any data collected through this method in the final sample. Consequently, we opted to exclude all such data because of their low validity.

A strength of this study is that, to the best of our knowledge, it is the first study to provide a direct comparison between values estimated from EQ-VT and DCEd data, within the same population, albeit not in the same sample, providing a reliable comparison between these methods. Furthermore, the results are robust under different exclusion thresholds for completion time. The effect of using different thresholds for exclusion has very little impact on the estimates values, suggesting that the choice of threshold does not affect the results of the study.

Conclusions

This study showed excellent agreement between DCEd and EQ-VT value sets, be it cTTO-only-based value sets, DCE value sets anchored on the cTTO data, or hybrid models. It is important to emphasize that these results were shown to be contingent upon the accommodation of nonlinear time preferences in the DCEd method. If linear time preferences are imposed on the DCEd data, estimated values differ substantially from EQ-VT-derived values. The findings of this study suggest that DCEd with accommodation of nonlinear time preferences provides a feasible alternative for valuation studies, providing a cheaper, logistically less challenging alternative to EQ-VT, which produces similar results.

Author Disclosures

Author disclosure forms can be accessed below in the Supplemental Material section.

Supplemental Material

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REFERENCES

1. Drummond MF, Sculpher MJ, Claxton K, Stoddart GL, Torrance GW. *Methods for the Economic Evaluation of Health Care Programmes*. Oxford, United Kingdom: Oxford university press; 2015.
2. Herdman M, Gudex C, Lloyd A, et al. Development and preliminary testing of the new five-level version of EQ-5D (EQ-5D-5L). *Qual life Res*. 2011;20(10):1727–1736.
3. Brazier J, Roberts J, Deverill M. The estimation of a preference-based measure of health from the SF-36. *J Health Econ*. 2002;21(2):271–292.
4. Froberg DG, Kane RL. Methodology for measuring health-state preferences—II: scaling methods. *J Clin Epidemiol*. 1989;42(5):459–471.
5. Janssen BM, Oppe M, Versteegh MM, Stolk EA. Introducing the composite time trade-off: a test of feasibility and face validity. *Eur J Health Econ*. 2013;14(suppl 1):5–13.
6. Ryan M, Gerard K, Amaya-Amaya M, eds. *Using Discrete Choice Experiments to Value Health and Health Care*. 11. Berlin Germany: Springer Science and Business Media; 2007.
7. Stolk E, Ludwig K, Rand K, van Hout B, Ramos-Goñi JM. Overview, Update, and Lessons Learned From the International EQ-5D-5L Valuation Work: version 2 of the EQ-5D-5L valuation protocol. *Value Health*. 2019;22(1):23–30.
8. Oppe M, Rand-Hendriksen K, Shah K, Ramos-Goñi JM, Luo N. EuroQol protocols for time trade-off valuation of health outcomes. *Pharmacoeconomics*. 2016;34(10):993–1004.
9. Devlin N, Roudijk B, Ludwig K, eds. *Value Sets for EQ-5D-5L: A Compendium, Comparative Review & User Guide*. Cham, Switzerland: Springer; 2022.
10. Roudijk B, Ludwig K, Devlin N. EQ-5D-5L value set summaries. In: Devlin N, Roudijk B, Ludwig K, eds. *Value Sets for EQ-5D-5L: A Compendium, Comparative Review & User Guide*. Cham, Switzerland: Springer; 2022:55–212.
11. Omelyanovskiy V, Musina N, Ratushnyak S, et al. Valuation of the EQ-5D-3L in Russia. *Qual life Res*. 2021;30(7):1997–2007.
12. Chemli J, Drira C, Felfel H, et al. Valuing health-related quality of life using a hybrid approach: Tunisian value set for the EQ-5D-3L. *Qual life Res*. 2021;30(5):1445–1455.
13. Malik M, Gu NG, Hussain A, Roudijk B, Purba FD. The EQ-5D-3L valuation study in Pakistan. *Pharmacoecon Open*. 2023;7(6):963–974.
14. EuroQol Research Foundation. https://euroqol.org/eq-5d-instruments/eq-5d-5l-about/valuation-standard-value-sets/. Accessed August 14, 2023.
15. Jiang R, Shaw J, Mühlbacher A, et al. Comparison of online and face-to-face valuation of the EQ-5D-5L using composite time trade-off. *Qual life Res*. 2021;30(5):1433–1444.
16. Jonker MF, Roudijk B, Maas M. The sensitivity and specificity of repeated and dominant choice tasks in discrete choice experiments. *Value Health*. 2022;25(8):1381–1389.
17. Jonker MF. The garbage class mixed logit model: accounting for low-quality response patterns in discrete choice experiments. *Value Health*. 2022;25(11):1871–1877.
18. Bansback N, Hole AR, Mulhern B, Tsuchiya A. Testing a discrete choice experiment including duration to value health states for large descriptive systems: addressing design and sampling issues. *Soc Sci Med*. 2014;114(100):38–48.
19. Craig BM, Greiner W, Brown DS, Reeve BB. Valuation of child health-related quality of life in the United States. *Health Econ*. 2016;25(6):768–777.
20. Craig BM, Reeve BB, Brown PM, et al. US valuation of health outcomes measured using the PROMIS-29. *Value Health*. 2014;17(8):846–853.
21. Jonker MF, Attema AE, Donkers B, Stolk EA, Versteegh MM. Are health state valuations from the general public biased? A test of health state reference dependency using self-assessed health and an efficient discrete choice experiment. *Health Econ*. 2017;26(12):1534–1547.
22. Lim S, Jonker MF, Oppe M, Donkers B, Stolk E. Severity-stratified discrete choice experiment designs for health state evaluations. *Pharmacoeconomics*. 2018;36(11):1377–1389.
23. Jonker MF, Donkers B, de Bekker-Grob EW, Stolk EA. Advocating a paradigm shift in health-state valuations: the estimation of time-preference corrected QALY tariffs. *Value Health*. 2018;21(8):993–1001.
24. Craig BM, Rand K, Bailey H, Stalmeier PF. Quality-adjusted life-years without constant proportionality. *Value Health*. 2018;21(9):1124–1131.
25. Bailey H, Jonker MF, Pullenayegum E, et al. The EQ-5D-5L valuation study for Trinidad and Tobago. *Health Qual Life Outcomes*. 2024;22:51.
26. Oppe M, Norman R, Yang Z, van Hout B. Experimental design for the valuation of the EQ-5D-5L. In: Devlin N, Roudijk B, Ludwig K, eds. *Value Sets for EQ-5D-5L: A Compendium, Comparative Review & User Guide*. Cham, Switzerland: Springer; 2022:29–54.
27. Jonker MF, Donkers B, de Bekker-Grob E, Stolk EA. Attribute level overlap (and color coding) can reduce task complexity, improve choice consistency, and decrease the dropout rate in discrete choice experiments. *Health Econ*. 2019;28(3):350–363.
28. Jonker MF, Donkers B, de Bekker-Grob EW, Stolk EA. Effect of level overlap and color coding on attribute non-attendance in discrete choice experiments. *Value Health*. 2018;21(7):767–771.
29. Ramos-Goñi JM, Oppe M, Slaap B, Busschbach JJ, Stolk E. Quality control process for EQ-5D-5L valuation studies. *Value Health*. 2017;20(3):466–473.
30. Jonker MF, Norman R. Not all respondents use a multiplicative utility function in choice experiments for health state valuations, which should be reflected in the elicitation format (or statistical analysis). *Health Econ*. 2022;31(2):431–439.
31. Roudijk B, Sajjad A, Essers B, Lipman S, Stalmeier P, Finch AP. A value set for the EQ-5D-Y-3L in the Netherlands. *Pharmacoeconomics*. 2022;40(suppl 2):193–203.
32. Jonker MF, Bliemer MC. On the optimization of Bayesian D-efficient discrete choice experiment designs for the estimation of QALY tariffs that are corrected for nonlinear time preferences. *Value Health*. 2019;22(10):1162–1169.
33. Himmler S, Jonker M, van Krugten F, Hackert M, van Exel J, Brouwer W. Estimating an anchored utility tariff for the well-being of older people measure (WOOP) for the Netherlands. *Soc Sci Med*. 2022;301:114901.
34. Attema AE, Brouwer WB. The value of correcting values: influence and importance of correcting TTO scores for time preference. *Value Health*. 2010;13(8):879–884.
35. Dolan P, Stalmeier P. The validity of time trade-off values in calculating QALYs: constant proportional time trade-off versus the proportional heuristic. *J Health Econ*. 2003;22(3):445–458.
36. Lipman SA, Attema AE, Versteegh MM. Correcting for discounting and loss aversion in composite time trade-off. *Health Econ*. 2022;31(8):1633–1648.
37. Attema AE, Brouwer WB, Claxton K. Discounting in economic evaluations. *Pharmacoeconomics*. 2018;36(7):745–758.